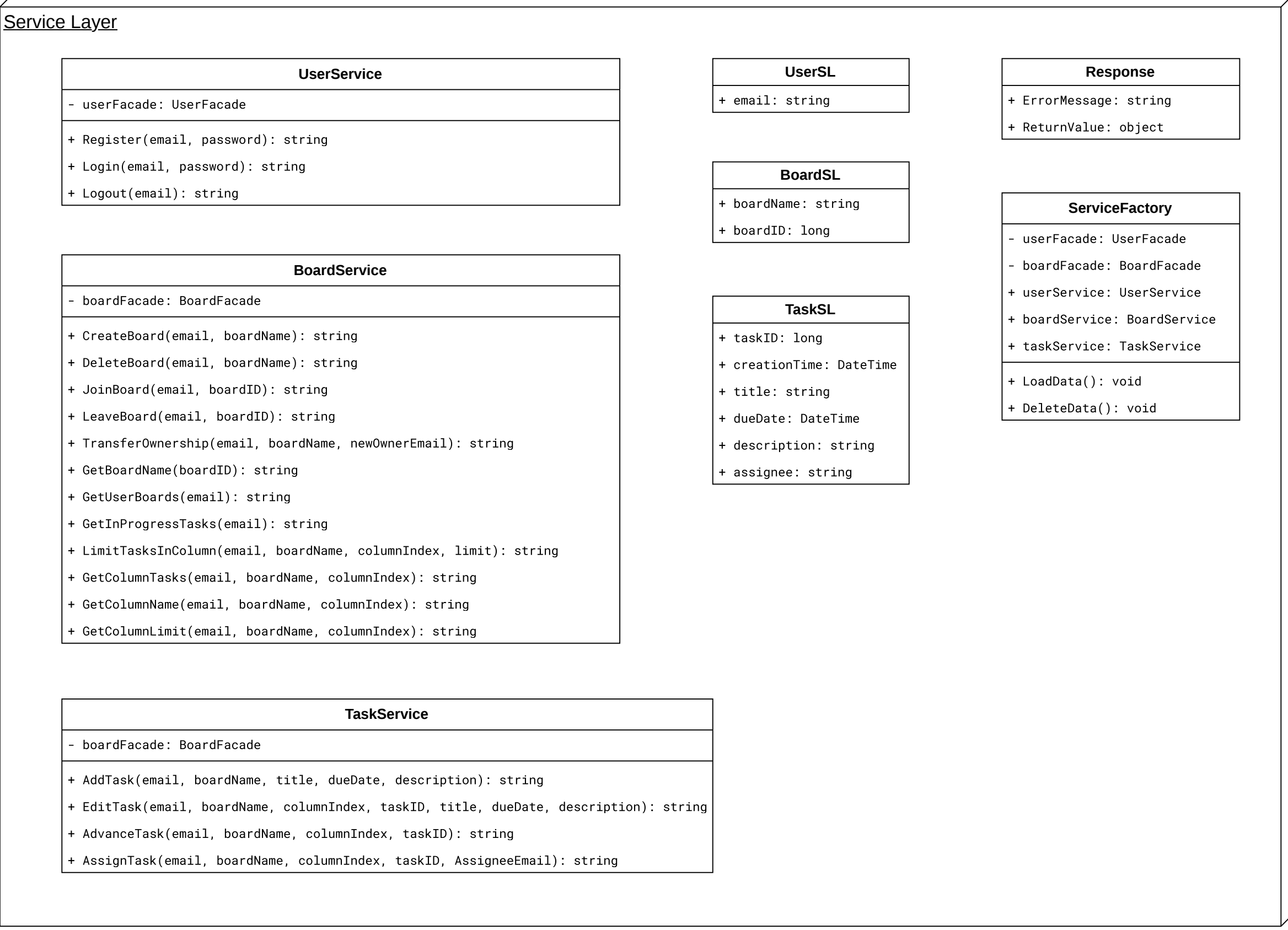




Response

+ ErrorMessage: string

+ ReturnValue: object

ServiceFactory

- userFacade: UserFacade

- boardFacade: BoardFacade

+ userService: UserService

+ boardService: BoardService

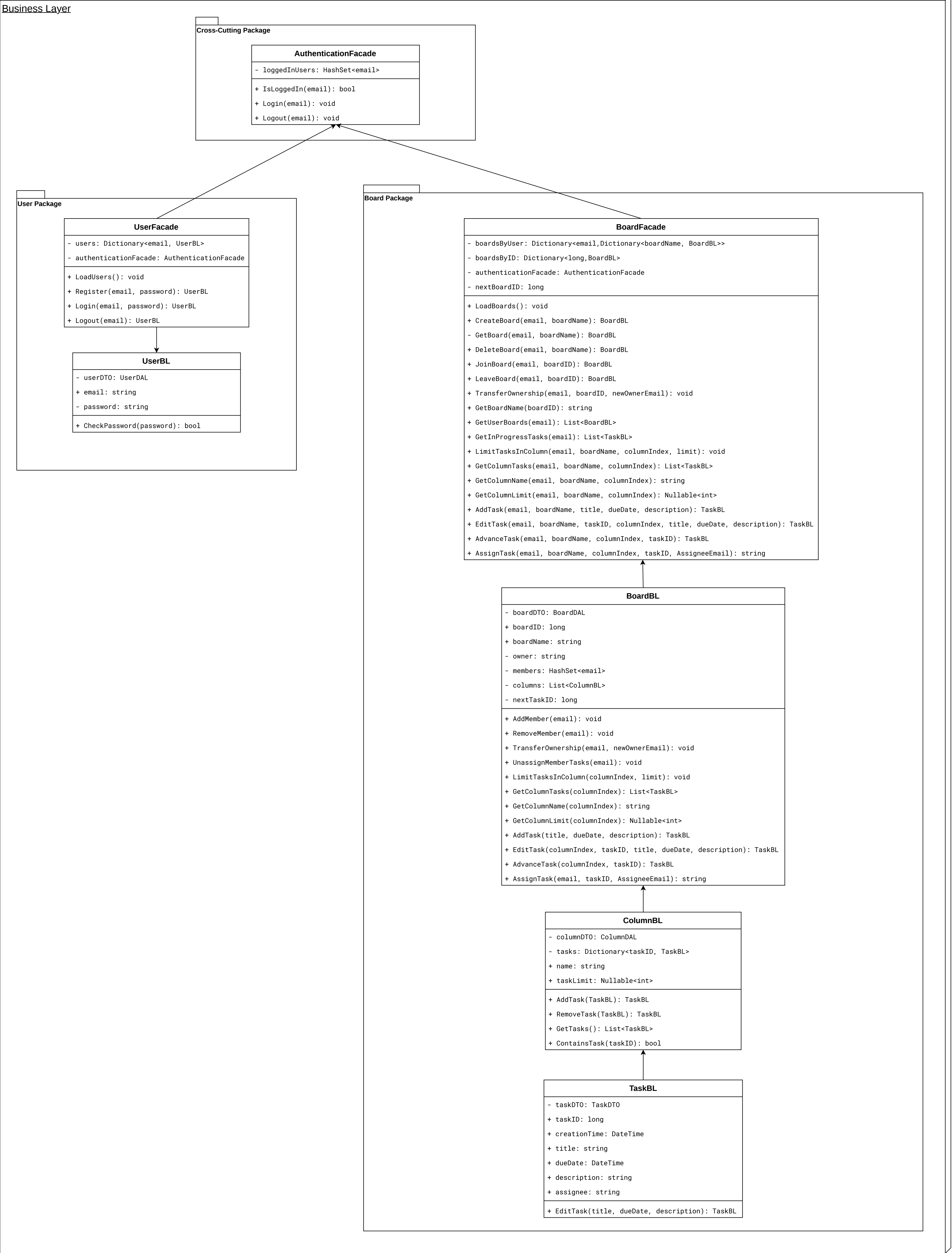
+ taskService: TaskService

+ LoadData(): void

+ DeleteData(): void

Changes

- Added a database and data access layer
- Added unique identifiers to boards
- Added the ability for users to join and leave boards created by other users
- Added a way for the owner of a board to transfer the ownership to a different user
- Added an "assignee" property to a task, and a way to control it
- Added methods to retrieve a board's name and retrieve all the boards belonging to a user



Board Package

BoardFacade

- boardsByUser: Dictionary<email, Dictionary<boardName, BoardBL>>

- boardsByID: Dictionary<long, BoardBL>

- authenticationFacade: AuthenticationFacade

- nextBoardID: long

+ LoadBoards(): void

+ CreateBoard(email, boardName): BoardBL

- GetBoard(email, boardName): BoardBL

+ DeleteBoard(email, boardName): BoardBL

+ JoinBoard(email, boardID): BoardBL

+ LeaveBoard(email, boardID): BoardBL

+ TransferOwnership(email, boardID, newOwnerEmail): void

+ GetBoardName(boardID): string

+ GetUserBoards(email): List<BoardBL>

+ GetInProgressTasks(email): List<TaskBL>

+ LimitTasksInColumn(email, boardName, columnIndex, limit): void

+ GetColumnTasks(email, boardName, columnIndex): List<TaskBL>

+ GetColumnName(email, boardName, columnIndex): string

+ GetColumnLimit(email, boardName, columnIndex): Nullable<int>

+ AddTask(email, boardName, title, dueDate, description): TaskBL

+ EditTask(email, boardName, taskID, columnIndex, title, dueDate, description): TaskBL

+ AdvanceTask(email, boardName, columnIndex, taskID): TaskBL

+ AssignTask(email, boardName, columnIndex, taskID, AssigneeEmail): string

BoardBL

- boardDTO: BoardDAL

+ boardID: long

+ boardName: string

- owner: string

- members: HashSet<email>

- columns: List<ColumnBL>

- nextTaskID: long

+ AddMember(email): void

+ RemoveMember(email): void

+ TransferOwnership(email, newOwnerEmail): void

+ UnassignMemberTasks(email): void

+ LimitTasksInColumn(columnIndex, limit): void

+ GetColumnTasks(columnIndex): List<TaskBL>

+ GetColumnName(columnIndex): string

+ GetColumnLimit(columnIndex): Nullable<int>

+ AddTask(title, dueDate, description): TaskBL

+ EditTask(columnIndex, taskID, title, dueDate, description): TaskBL

+ AdvanceTask(columnIndex, taskID): TaskBL

+ AssignTask(email, taskID, AssigneeEmail): string

ColumnBL

- columnDTO: ColumnDAL

- tasks: Dictionary<taskID, TaskBL>

+ name: string

+ taskLimit: Nullable<int>

+ AddTask(TaskBL): TaskBL

+ RemoveTask(TaskBL): TaskBL

+ GetTasks(): List<TaskBL>

+ ContainsTask(taskID): bool

TaskBL

- taskDTO: TaskDAL

+ taskID: long

+ creationTime: DateTime

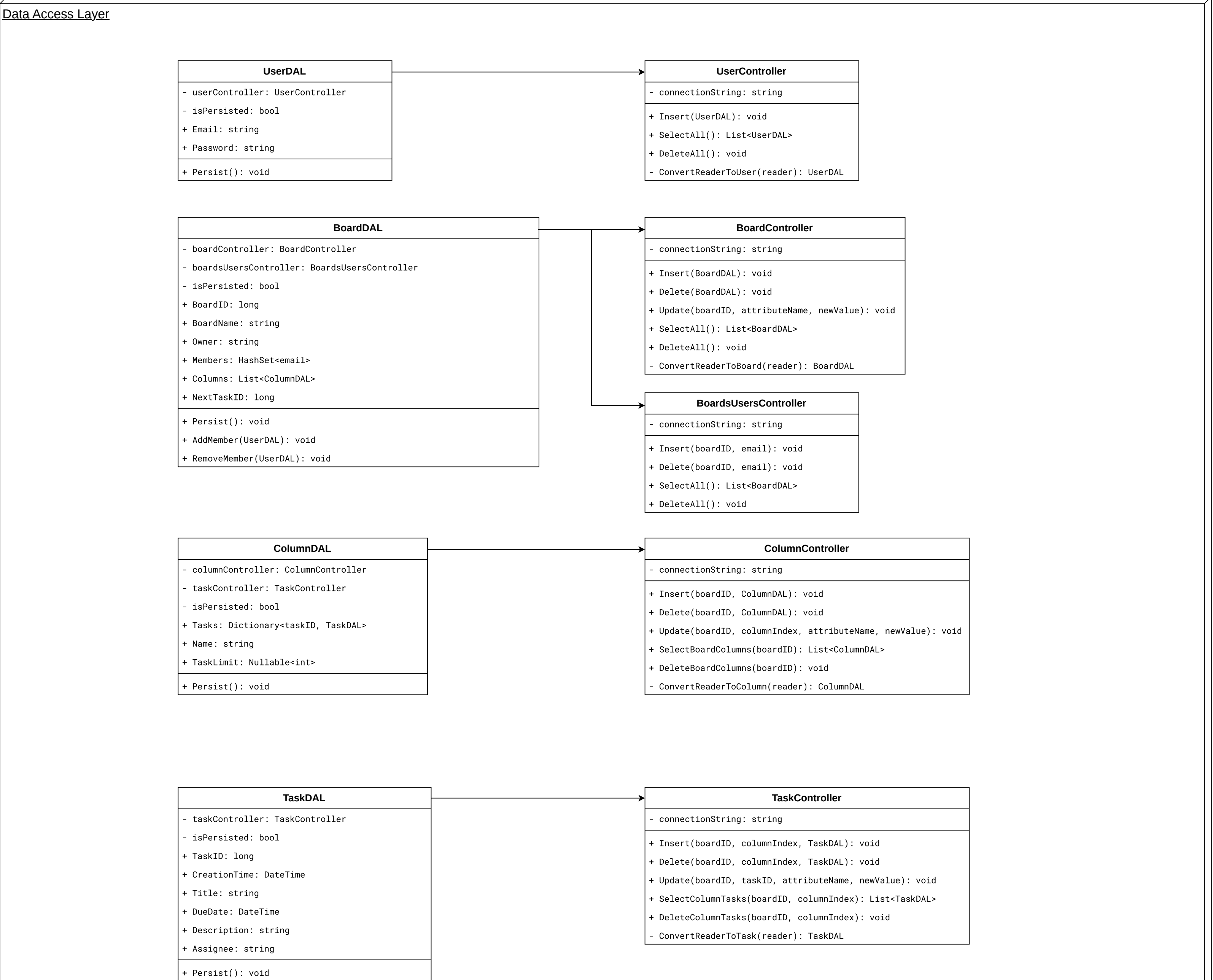
+ title: string

+ dueDate: DateTime

+ description: string

+ assignee: string

+ EditTask(title, dueDate, description): TaskBL



Database

